

Justin Jasper

(770) 480-8093 | jtjasper@stanford.edu | <https://www.linkedin.com/in/justin-jasper>

EDUCATION

Stanford University

Master of Science in Computer Science (Concentration: Artificial Intelligence)

Stanford, CA

June 2026

Stanford University

Bachelor of Science in Bioengineering

Stanford, CA

June 2025

EXPERIENCE

Centre for Artificial Intelligence and Robotics

Biomedical AI Intern

Hong Kong, China

June 2025 - September 2025

- Developed and deployed a novel full-stack landmark detection model for cardiac ultrasound images to assist non-specialist physicians in accurate diagnosis (Pytorch, Python, React, PostgreSQL).
- Collaborated with the Emergency Department at Prince of Wales Hospital to test the cardiac ultrasound model and integrate it into clinical workflows for point-of-care ultrasound
- Designed a hallucination-resistant LLM architecture leveraging biomedical knowledge graphs to support clinical decision-making and treatment plan formulation based on patient medical history and clinical data (Python, Neo4j)

MITRE Corporation

Computational Biology Intern

Mclean, VA

June 2024 - December 2024

- Designed automation software to optimize high-throughput training and fine-tuning processes for open-source large language models (LLMs).
- Developed an LLM-driven retrieval augmented generation system to extract and summarize key insights from complex scientific literature, streamlining knowledge access for synthetic biology workflows (Used scientific python, LangChain, LlamaIndex, Kubernetes).
- Constructed the BioNet Domain-Specific Language, an open-source programming language designed to enable extensive scaling of synthetic biology experiments and industrial workflows.

Omniwear AI

CoFounder, CEO

Palo Alto, CA

December 2022 - December 2023

- Led the development of a cutting-edge B2B virtual try-on platform for e-commerce powered by computer vision and generative AI, catering specifically to the D2C retail clothing sector.
- Spearheaded the B2B growth strategy, successfully acquiring early customer brands. Notably, secured partnerships with a retail giant valued at over \$1 billion and a retailer valued at around \$250 million.
- Established and maintained direct communication channels with corporate executives, ensuring seamless collaboration and addressing clients' unique needs and concerns.
- Participated in Launchpad (Stanford's premier graduate accelerator) Spring '23 and Entrepreneur First's NYC unicorn training program.

PROJECTS

CV Image Processing for CT and Ultrasound-based Tumor Detection

- Developed a computer-vision (CV)-based image processing pipeline for lung CT scans using Python and OpenCV to segment and reconstruct 3D lung images for clinical diagnosis of suspected cancer.
- Explored deep learning for tumor segmentation using an Attention U-Net model with TensorFlow, achieving high accuracy in classifying malignant and benign lesions.
- Implemented K-means clustering to classify lung tissue and air regions, achieving accurate segmentation of lung abnormalities.

Python-Based Probabilistic Modeling of Biological Systems

- Developed Python-based computational models to simulate and analyze molecular biological systems, including ion channel activation, protein-ligand binding, and polymer conformations, using Boltzmann statistical mechanics.

- Applied probabilistic modeling techniques to predict multi-state biological behaviors, enabling insights into molecular dynamics and system energetics.

Computational Analysis of NMDA Receptor Antagonist Binding for Chronic Pain Therapeutics

- Investigated the molecular interactions between NMDA receptor antagonists and their binding sites using molecular docking simulations with SwissDock and AutoDock Vina.
- Designed and optimized docking workflows, refining model parameters to maximize accuracy in predicting ligand-receptor interactions.
- Quantified binding affinities to assess antagonist efficacy, providing computational insights for drug repurposing in chronic pain management.

Synthetic Biosensor Development for Pathogen Detection

- Developed a novel biosensor to detect waterborne *Salmonella enterica* using a SynNotch receptor system with an anti-*Salmonella* single-chain variable fragment (scFv) transfected into human embryonic kidney cells (HEK293)
- Designed and transfected mammalian cell cultures (HEK293) with receptor plasmids, optimizing expression through flow cytometry and Western cell blotting analysis.
- Leveraged common wet lab techniques including polymerase chain reaction (PCR), gel electrophoresis, mammalian cell cultures, and bacterial cultures

Computer Vision for Meal Nutritional Analysis

- Developed a computer vision (CV)-based web application that detects and classifies food items from user-uploaded images.
- Trained a ResNet18 convolutional neural network model on a food image dataset from Kaggle (using Tensorflow)
- Implemented an object detection pipeline to localize and identify food items before classification.
- Built a full-stack web interface using HTML (frontend) and a Python Flask backend to allow users to upload images for real-time food classification.
- <https://github.com/justinjasper/cv-nutritional-analysis>

SKILLS

- Proficient in Python (NumPy, Pandas, SciKit-learn, TensorFlow), C++, C, MATLAB, git, UNIX, LLM training & deployment (LangChain, LlamaIndex, Kubernetes), retrieval augmented generation (RAG), neural networks
- Proficient in Spanish, with strong cross-cultural communication skills.